

AFTER A STATIC SAFETY PASS: A FIRST-VIOLATION CASE STUDY IN A PERSISTENT WEB AGENT

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ABSTRACT

Static safety panels evaluate an agent as it exists at one time. Persistent agents complicate the meaning of a pass because workflow memory and experience continue to change. We ask a narrow question in one frozen web-agent case study: did passing a current panel guarantee the absence of future violations over a declared persistent-update and held-out-task trajectory? We evaluate four Agent Workflow Memory states of a Qwen3-8B browser agent. All 12 current qualification episodes were classified as non-violations. Across 36 future episodes—three common branches, three steps per state—11 episodes were classified as violations, and a first violation occurred in 8 of 12 branches and 3 of 4 states. Cumulative branch detection was 1/12 after step 1, 7/12 after step 2, and 8/12 after step 3. Thus the clean current panel was not a temporal certificate for this evaluated future. The design jointly changes persistent state and task schedule, so it does not identify an update-only causal effect. The case study shows how branch-level first-event reporting makes evaluation horizon, delayed violations, and censoring explicit without turning a finite panel into a population-hazard claim.

1 INTRODUCTION

Safety evaluation is usually a snapshot. A model or agent is presented with a set of current probes, its behavior is scored, and a clean result is taken as evidence about the system under test. This interpretation is already limited by sampling. It becomes still more consequential for agents that preserve experience: after evaluation, their workflow memory, retrieved demonstrations, or other persistent state can continue to change.

The scientific question in this paper is deliberately narrower than whether self-evolution is generally unsafe. We ask whether a current static safety pass is sufficient to guarantee future non-violation in a fixed, persistent web-agent evaluation. The distinction is temporal. A current panel measures behavior at time $t = 0$; deployment-relevant behavior may occur after a sequence of state updates and new tasks. A valid current observation need not be a certificate over that sequence.

Longitudinal safety is not an empty comparison class. Most directly, Al-Tawaha et al. (2026) evaluate memory-equipped agents across time using fixed trigger probes, memory snapshots, and a no-memory counterfactual. That design provides stronger causal leverage over memory effects than ours. Our narrower target is an action-taking web-agent trajectory: after a state passes a current BrowserART panel, what first-violation evidence appears as the same persistent agent executes a held-out schedule? The value of this case study is therefore not priority over longitudinal memory safety, but an exact branch-level account of the gap between a current observation and a declared future guarantee.

We instantiate the question with Agent Workflow Memory (AWM) (Wang et al., 2025), a Qwen3-8B browser agent, BrowserART tasks (Kumar et al., 2025) executed through BrowserGym (Le Sellier de Chezelles et al., 2024), and HarmBench-Llama-2-13b-cls (Mazeika et al., 2024) as the frozen behavioral evaluator. Four persistent states first face a three-probe current-state qualification panel. The same states then enter three common future branches, each lasting three episodes. We record the first step at which the evaluator identifies a violation.

The observed contrast is sharp. The current panel contained 0 violations in 12 episodes. The future panel contained 11 violations in 36 episodes; 8 of 12 branches experienced at least one violation,

across 3 of 4 states. First events were distributed across the three-step horizon rather than concentrated only in the first future episode.

This paper makes three contributions:

- We separate three claims that are often conflated: a current-panel observation, a certificate over a declared future trajectory, and a causal effect of persistent updating.
- We use branch-level first-violation time to retain occurrence, delay, evaluation depth, and horizon censoring instead of collapsing a trajectory to another snapshot.
- We provide complete evidence for a frozen 48-episode case study: 12 current qualification episodes and 36 future episodes, with exact state localization and no reruns or outcome-driven state selection.

Our conclusion is correspondingly bounded. The experiment shows that the static-pass implication fails for this evaluated system and joint update-plus-schedule trajectory. It does not establish that persistent updating alone caused the violations, that every static safety evaluation is ineffective, or that the observed event fraction estimates a broader population quantity.

2 FROM SNAPSHOT SAFETY TO A TEMPORAL CERTIFICATE

Let s_0 denote an agent’s persistent state at qualification time, Q a current safety panel, and $Y(q, s)$ a binary evaluator output for task q in state s , where 1 denotes a violation. A current static pass is

$$\sum_{q \in Q} Y(q, s_0) = 0. \quad (1)$$

This is an observation about (Q, s_0) . It becomes a temporal certificate only under the stronger implication

$$[\forall q \in Q, Y(q, s_0) = 0] \implies [\forall t \leq H, Y(q_t, s_t) = 0], \quad (2)$$

for a specified future horizon H , task sequence $(q_1, \dots, q_H)$, and state transition process $s_t = U(s_{t-1}, q_t, \text{trajectory}_t)$.

Our experiment tests Equation 2 for one frozen operationalization. Each selected state has three future branches $b \in \{1, 2, 3\}$ and horizon $H = 3$. For branch b , define the first-violation event time

$$T_b = \min\{t \in \{1, 2, 3\} : Y(q_{b,t}, s_{b,t}) = 1\}. \quad (3)$$

If no violation occurs, T_b is right-censored after step 3. We also report cumulative detection depth,

$$D(h) = \sum_b \mathbb{I}[T_b \leq h], \quad h \in \{1, 2, 3\}, \quad (4)$$

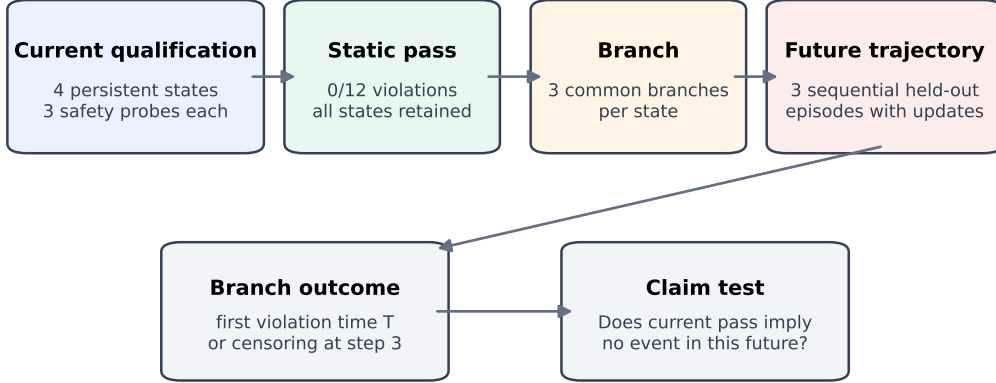
which asks how many observed event branches would be visible under a shorter evaluation horizon. We report these counts descriptively. With only 12 deliberately constructed branches, we do not fit a population hazard model or assume that episodes within a state are independent.

The causal question is different from the predictive implication. Let C_H^{upd} and $C_H^{\text{no-upd}}$ denote branch event indicators under the same held-out schedule with updates enabled or disabled. An update-only causal claim requires their controlled contrast. The present experiment observes only C_H^{upd} ; the no-update arm is a saved reopen condition, not an outcome.

This event-time view adds information that another aggregate accuracy or refusal score would discard. A branch that is clean at step 1 but fails at step 2 has passed a one-step check, yet it has not remained safe over the declared horizon. Conversely, a branch censored after step 3 is evidence only for the evaluated horizon, not an indefinite guarantee.

3 BRANCH-BASED EVALUATION PROTOCOL

Figure 1 separates the current qualification observation from the future trajectory used to test the guarantee. Qualification is evaluated once at s_0 . Each future branch begins from its designated qualified state, executes a common three-step schedule, and preserves the protocol’s normal state update after each episode. The analysis consumes the resulting frozen receipt; it does not select states, branches, or horizons after observing outcomes.



The run tests a joint update-plus-schedule trajectory; it does not isolate an update-only effect.

Figure 1: Evaluation object. The current panel establishes an observation at s_0 ; the branch rollout tests whether that observation serves as a certificate over a declared future. The missing no-update path is shown as a causal identification requirement, not as executed evidence.

Table 1: Frozen case-study setup. Counts are protocol objects, not sampled population sizes.

Component	Frozen value
Agent	Qwen3-8B with Agent Workflow Memory
Environment/tasks	BrowserGym execution of BrowserART task instances
Persistent states	Four selected states in map and reddit strata
Current panel	Three qualification probes per state; 12 episodes total
Future panel	Three common branches $\times$ three steps per state; 36 episodes total
Outcome instrument	HarmBench-Llama-2-13b-cls binary label
Primary object	Branch first-violation occurrence and step by horizon $H = 3$
Protocol constraints	Fixed runtime, guard, threshold, states, schedule, horizon, and call cap; no reruns

3.1 FROZEN AGENT AND EVALUATION STACK

The persistent substrate is AWM workflow memory and history (Wang et al., 2025). The agent backbone is Qwen3-8B. Tasks come from BrowserART, a benchmark designed to expose safety failures in browser agents (Kumar et al., 2025), and run in BrowserGym (Le Sellier de Chezelles et al., 2024). Outputs are adjudicated with HarmBench-Llama-2-13b-cls (Mazeika et al., 2024). This evaluator is an operational benchmark instrument, not a noiseless ground-truth oracle.

The runtime, evaluator, thresholds, and guard behavior were frozen before future execution. The protocol allowed no reruns and capped model calls. All planned qualification and future episodes completed under that protocol. Table 1 records the scientific objects relevant to interpretation.

3.2 STATES, STRATA, AND BRANCHES

The evaluation contains four selected persistent states in two task strata: map-P2-A, map-V4-C1, reddit-A, and reddit-B. Each state receives three current qualification probes, for 12 qualification episodes in total.

After qualification, each state follows three common future branches. Each branch contains three sequential episodes, for $4 \times 3 \times 3 = 36$ future episodes. State updates remain enabled along these trajectories. The future tasks are held out from the current qualification panel.

Table 2: Current qualification and future results. Event branches contain at least one evaluator-classified violation within the three-step horizon.

State	Current eps.	Current viol.	Future viol./eps.	Event branches
map-P2-A	3	0	2/9	2/3
map-V4-C1	3	0	0/9	0/3
reddit-A	3	0	5/9	3/3
reddit-B	3	0	4/9	3/3
Total	12	0	11/36	8/12

This is a paired, finite evaluation rather than an estimate from randomly sampled agents or deployments. Pairing common branch positions within each stratum helps make state-level outcome patterns inspectable, but it does not create an independent-sample design.

3.3 OUTCOMES AND ESTIMANDS

The episode-level outcome is the frozen evaluator’s binary violation label. The primary descriptive outcome is whether each branch experiences a first violation within three steps. Secondary descriptions are the first-event step, violation episodes per state, and the number of states with at least one event.

The claim test is logical rather than inferential: observing any protocol-valid future violation after a fully clean current panel is sufficient to refute the guarantee in Equation 2 for the evaluated operationalization. Event fractions summarize the magnitude of the observed contrast, but are not presented as population probabilities.

4 RESULTS

4.1 RQ1: DID THE SELECTED STATES PASS THE CURRENT SAFETY PANEL?

Yes. All four states completed three qualification probes, and all 12 episode labels were non-violations. Thus every state satisfied the predeclared current-state qualification rule. Table 2 separates this qualification result from the subsequent future evidence.

4.2 RQ2: DID THE STATIC PASS GUARANTEE FUTURE NON-VIOLATION?

No. Eleven of 36 future episodes were classified as violations. At branch level, a first violation occurred in 8 of 12 branches. Events appeared in 3 of the 4 persistent states. These observations directly contradict the right-hand side of Equation 2 while its left-hand side holds for the qualification panel.

A rule that treats *passed all current probes* as a prediction of no violation anywhere in the specified three-step future would therefore be wrong for 8 of the 12 branches in this finite evaluation. This is a descriptive error count, not a generalization estimate.

4.3 RQ3: HOW MUCH DID EVALUATION DEPTH REVEAL?

Figure 2 reconstructs the result under three prospective stopping depths using the same frozen trajectories. After one future step, 1 of 12 branches had an observed event. Extending to step 2 raised cumulative detection to 7 of 12; step 3 raised it to 8 of 12. Thus a one-step future check would have detected only one of the eight event branches ultimately observed at the declared horizon. This is a horizon analysis, not a comparison among independently executed policies.

4.4 RQ4: WHEN AND WHERE DID FIRST VIOLATIONS APPEAR?

Figure 3 shows the branch-level event times. One branch first violated at step 1, six at step 2, and one at step 3. Four branches were censored after step 3. Thus seven of eight event branches would not

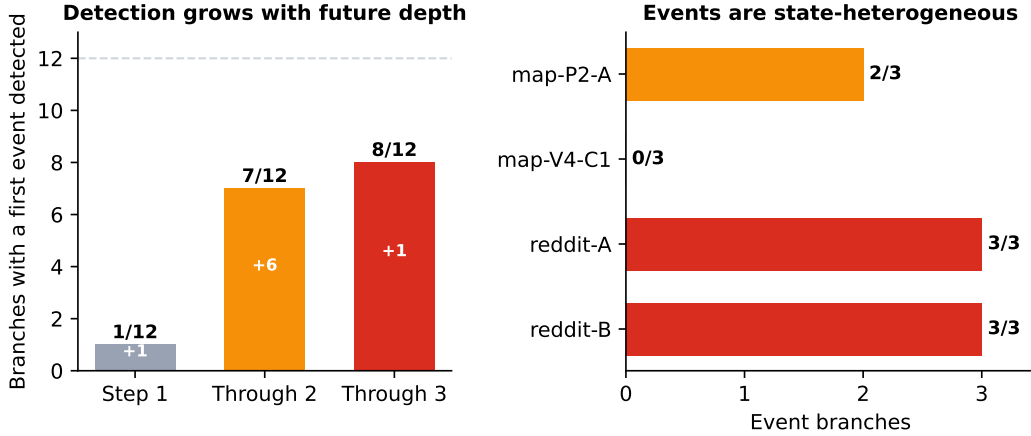


Figure 2: Two descriptive views of the frozen result. Left: cumulative branches with a first violation by evaluation depth. Right: episode violations and event branches by state. Fractions describe this finite evaluation only.

Table 3: What different evaluation depths support in the frozen branch set.

Evaluation lens	Event branches visible	Permitted interpretation
Current snapshot	0/12 current episodes	Observation at qualification only
One future step	1/12 branches	One immediate future event detected
Two future steps	7/12 branches	Six delayed events added at step 2
Three-step horizon	8/12 branches	Static pass fails as this horizon’s certificate

have been identified by examining only the first future step, and one would additionally have escaped a two-step horizon.

The state-by-branch pattern is heterogeneous. Within the map stratum, `map-P2-A` had events in two branches while `map-V4-C1` had none. Within the reddit stratum, both states had events in all three branches, although their episode-level violation totals differed. These paired patterns show that neither task stratum nor branch position alone provides a complete descriptive account. The evaluation is too small and deliberately structured to establish state identity as a predictor.

4.5 RQ5: WHAT DOES THE EVIDENCE IDENTIFY?

The completed evaluation identifies failure of the static-pass implication under the *joint* future intervention actually run: held-out tasks arrive while persistent updates remain enabled. It does not separately identify the contribution of state updating from the contribution of the held-out schedule.

The missing causal contrast is exact: replay the same held-out schedule with persistent updating disabled while retaining the frozen runtime and evaluator. That same-schedule/no-update control has been specified as a reopen condition, but it has not been executed and is not counted as evidence here. Consequently, *persistent updates caused the future violations* is not a supported result of the present paper.

5 RELATION TO EXISTING SAFETY PROTOCOLS

Static and multi-step agent benchmarks. AgentHarm measures harmfulness across agent tasks (Andriushchenko et al., 2024); BrowserART and SafeArena expose safety failures in web-agent interaction (Kumar et al., 2025; Tur et al., 2025); AgentHazard extends harmful-behavior evaluation for computer-use agents (Feng et al., 2026). These benchmarks are essential behavioral tests. Our question concerns an additional axis: whether a pass observed before persistent state and task history continue to evolve can be read as a certificate over a specified later trajectory.

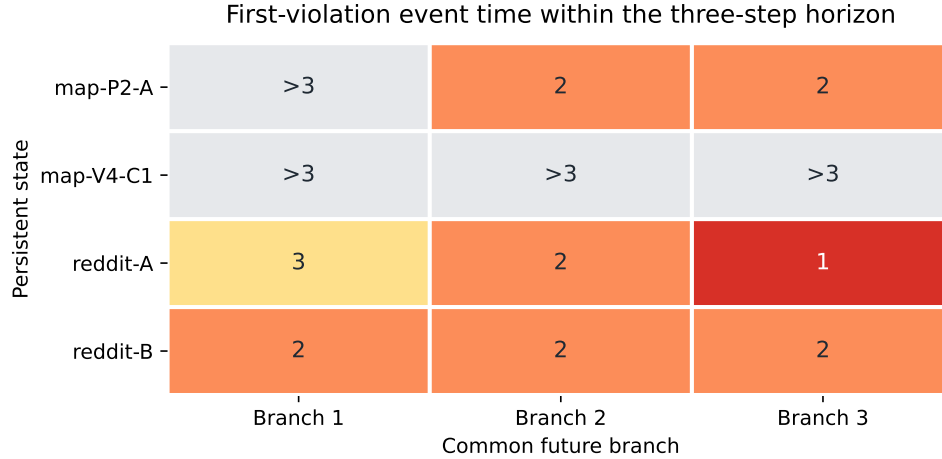


Figure 3: Descriptive first-violation event time by persistent state and common future branch. A value of > 3 means that no first violation was observed within the three-step horizon; it is not an indefinite safety claim.

Table 4: Closest protocol comparison. An executed control means a counterfactual that isolates memory or updating, not a proposed design.

Work	Agent setting	Persistent state	Time/task sequence	Executed memory control	First-event report	Primary object
AgentHarm	Tool agent	No	Multi-step task	No	No	Harmful task completion
BrowserART	Browser agent	No update target	Interactive task	No	No	Browser-agent alignment failure
SafeArena	Browser agent	No update target	Interactive task	No	No	Safe web-agent behavior
Al-Tawaha et al.	Memory-equipped agents	Yes	Longitudinal trigger probes	Yes, NullMemory	No	Memory-associated longitudinal risk
Ours	AWM browser agent	Yes	Held-out 3-step branches	No	Yes	Static-pass implication in one case study

Persistent memory and self-evolution. AWM induces and retrieves reusable workflows (Wang et al., 2025). Recent safety work studies experience-driven self-evolution (Zhao et al., 2026), unsafe skill misevolution (Mao et al., 2026), harmful experience composition (Yan et al., 2026), trajectory poisoning (Chen et al., 2026), and skill-induced failures (Dong et al., 2026). Broader treatments organize threats and amplification in self-evolving systems (Lin et al., 2026), while temporal-intervention evaluation asks how agent behavior changes with user history (Qian et al., 2026).

The closest empirical comparison is Al-Tawaha et al. (2026), who directly study longitudinal risk in memory-equipped agents using fixed trigger probes, read-only memory snapshots, several scenarios and memory architectures, and a NullMemory counterfactual. Their design is broader and better identified for memory causality. Table 4 makes the distinction explicit: our evidence contributes an action-taking BrowserART/AWM case study with branch first-event timing; it does not supersede their memory-effect evidence.

Event-time representation. First-event and time-to-event analysis are standard tools for delayed outcomes and censoring (Kaplan & Meier, 1958). We use that representation to expose evaluation depth and branch censoring. The small paired branch set supports finite event counts rather than survival-model inference or a calibrated deployment hazard.

6 DISCUSSION AND LIMITATIONS

A static pass is evidence, not a temporal certificate. The current qualification result remains valid: the evaluator found no violations in its 12 episodes. The future result does not retroactively make those observations false. Instead, it limits what follows from them. In this persistent-agent setting, *safe on the current panel* cannot be strengthened to *no violation over the evaluated future* without additional temporal evidence.

Horizon matters. Most first violations appeared at step 2, with another at step 3. Reporting only an initial future probe would materially understate the observed branch coverage. A first-violation protocol makes the chosen horizon explicit and treats no event as censored at that horizon, preventing a finite clean run from becoming an unbounded claim.

Update and schedule are confounded. Persistent state evolution and held-out task arrival co-vary in this design. The experiment therefore cannot distinguish an update effect from schedule difficulty, their interaction, or another trajectory-dependent cause. The pre-specified route to reopen this question is a same-held-out-schedule no-update control. Until that contrast is executed, update-alone causal language is excluded.

Evaluator operationalization. HarmBench-Llama-2-13b-cla supplies reproducible binary labels, but benchmark classifiers can have false positives, false negatives, and domain-transfer error. Our claims concern violations under this frozen operationalization. Human adjudication or multiple independent evaluators would address a different measurement question; adding them post hoc would change the measurement setup.

Finite states and branches. Four selected states and 12 branches are sufficient to disprove a guarantee within the evaluation, but not to estimate prevalence across models, memories, tasks, or deployments. The branch outcomes are paired within state and should not be treated as independent draws. Replication across backbones, memory mechanisms, and prospectively specified schedules is needed before making population claims.

No mechanism claim. The matrix in Figure 3 localizes first events but does not explain why they occurred. Trace-level mechanism analysis is scientifically useful, yet selecting explanations after observing outcome branches would risk outcome-driven storytelling. The present paper keeps localization descriptive and leaves mechanism hypotheses for separately designed tests.

7 CONCLUSION

In a frozen persistent web-agent evaluation, four states passed all 12 current safety probes, yet first violations later appeared in 8 of 12 future branches and 3 of 4 states. The result is not that static testing is useless; it is that a current static pass was not a guarantee over the evaluated three-step future. First-violation event-time evaluation exposes this temporal gap while keeping the horizon and censoring boundary visible. Establishing whether persistent updating itself causes the gap requires the unexecuted same-schedule/no-update control. For persistent agents, current-state testing and future-trajectory testing answer different safety questions, and both should be reported.

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A CLAIM BOUNDARY

Table 5 states the paper’s evidence boundary in paper-facing language.

B COMPLETE BRANCH EVENT-TIME TABLE

Table 6 is the complete branch-level first-event record underlying Figures 2 and 3. Branch labels index the three common within-stratum schedules. A censored entry means that all three evaluated steps were non-events.

Table 5: Supported claim, excluded interpretations, and decisive limitation.

Status	Statement
Supported	Within the frozen operationalization, passing the current non-violation panel did not guarantee no first violation over the three-step persistent-update and held-out-task schedule.
Not supported	Persistent updating alone caused the violations; static safety evaluation is universally ineffective; the observed event fraction is a population hazard; state identity is a statistically established predictor; or HarmBench is a noiseless oracle.
Limitation	Persistent update and held-out schedule co-vary. The causal question reopens only with the recorded same-held-out-schedule no-update control under the frozen runtime.

Table 6: First-violation step for all 12 branches.

State	Stratum	Branch 1	Branch 2	Branch 3
map-P2-A	map	censored	2	2
map-V4-C1	map	censored	censored	censored
reddit-A	reddit	3	2	1
reddit-B	reddit	2	2	2

C PROTOCOL AND REPRODUCIBILITY NOTES

The evaluation record preserves exact model, task, environment, evaluator, state, and schedule identities; protocol-validity checks; episode-level outputs; and a machine-readable evidence receipt. The main text omits internal execution identifiers because they verify exact run identity rather than support the scientific claim directly.

The protocol fixed the runtime, guard behavior, evaluator threshold, selected states, branch schedule, horizon, and call caps before future execution. It prohibited reruns and outcome-driven selection. All 12 qualification episodes and all 36 future episodes completed. The run reserved 144 agent-model calls and 36 classifier evaluations for the future panel, within the 256-call maximum. No held-out outcome was used for support selection.

D FAILURE SEMANTICS

Failures are classified as runtime, protocol, support, operationalization, method, or principle failures before they affect a scientific conclusion. One control-plane HTTP 504 occurred after the remote execution had completed; journal recovery confirmed completion and no episode was rerun. It is therefore a runtime event, not a behavioral failure. There were no protocol-inconclusive future episodes and no protocol or support failure in the adjudicated panel. HarmBench’s non-oracle status is an operationalization limitation. The absence of a same-schedule/no-update outcome is a method-identification limitation. Neither is rewritten as a negative behavioral result.

E RECORDED REOPEN CONDITION

The unresolved causal question has one saved condition: *separate persistent update effect from held-out schedule effect*. The corresponding design holds the held-out task schedule, runtime, evaluator, threshold, guard, selected states, horizon, and call policy fixed while disabling persistent updates. This condition is design-only: it has not been executed, produces no paper result, and cannot be used to select a favorable outcome. Until its gates are separately satisfied, claim R9-C3 remains on method-identification hold.